ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · A1 · BODY SYSTEMS

Organs and their jobs

★☆☆
Grade 5

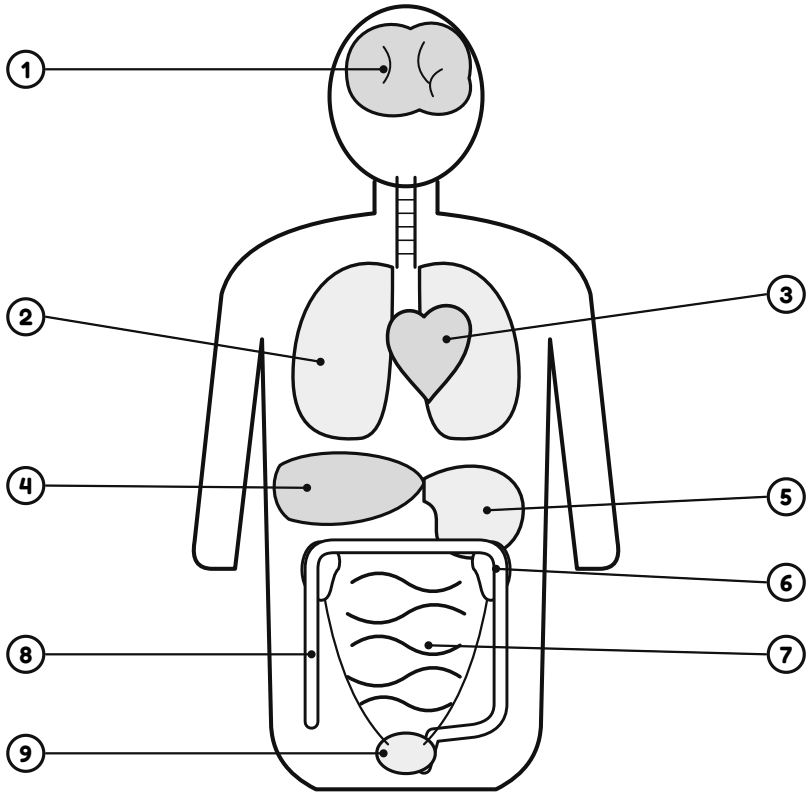
Learning goals

- Label the major organs.
- Match organs to systems and jobs.

1 **Label the organs.** [9]

LABELS

liver **small intestine** **stomach** **brain** **bladder** **lungs** **heart** **kidneys** **large intestine**



1 brain	2 lungs	3 heart
4 liver	5 stomach	6 kidneys
7 small intestine	8 large intestine	9 bladder

2 Write the organ system each organ belongs to. [6]

Organ	Organ system
heart	circulatory
lungs	respiratory
stomach	digestive
kidneys	excretory
brain	nervous
skull	skeletal

3 Match each organ to its job. [6]

Draw a line to match each pair, or write the letter.

heart ●	→ F	● **A** nutrients pass into the blood
lungs ●	→ E	● **B** filter blood and make urine
small intestine ●	→ A	● **C** protect the heart and lungs
kidneys ●	→ B	● **D** controls the body and processes messages
brain ●	→ D	● **E** take in oxygen and remove carbon dioxide
ribs ●	→ C	● **F** pumps blood around the body

4 Fill in the gaps. [4]

WORD BANK

blood vessels **bladder** **stomach** **diaphragm** **pulse** **nerve**

Blood is carried in tubes called blood vessels . The beat you can feel in your wrist is your pulse . Urine is stored in the bladder . The diaphragm is a muscle that helps us breathe.

ANSWER KEY

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THINK IT THROUGH · A1 · BODY SYSTEMS

How the systems work together

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Grade 5

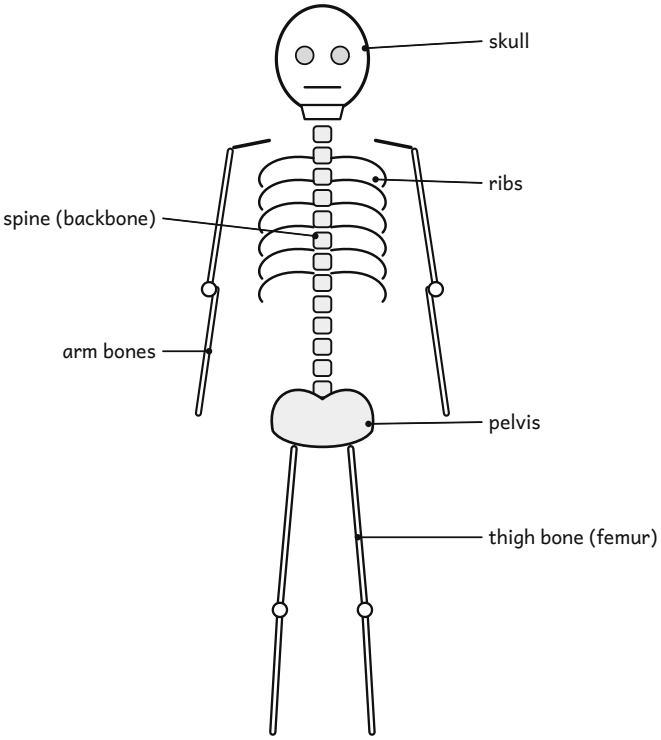
Learning goals

- Order the journey of food through the body.
- Explain links between organ systems.
- Interpret pulse data.

1 **Number the stages of digestion from 1 to 6.** [3]

5	Water is taken back in the large intestine.
1	Teeth chew food in the mouth.
6	Waste leaves the body.
4	Nutrients pass into the blood in the small intestine.
2	Food travels down the oesophagus.
3	The stomach churns food with acid.

2 **Which bones protect which organs? Complete the table.** [3]



Bone	Organ it protects
skull	brain
ribs	heart and lungs
spine (backbone)	spinal cord

3 **Explain why the circulatory system and the respiratory system must work together.** [2]

The lungs take oxygen into the blood; the heart pumps that blood to all the cells. Blood brings carbon dioxide back to the lungs to be breathed out.

4 **A boy touches a hot pan and quickly pulls his hand away. Describe the path of the message.** [2]

Nerves in the skin of his hand send a message to the spinal cord/brain; a message is sent back along nerves to the arm muscles, which move the hand away.

5 **Why do you breathe faster and your heart beat faster when you exercise?** [2]

Working muscles need more oxygen and energy; faster breathing takes in more oxygen and a faster heartbeat carries it to the muscles more quickly (and removes more carbon dioxide).

ANSWER KEY

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INVESTIGATE · A1 · BODY SYSTEMS

How does exercise change my pulse?

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Grade 5

Learning goals

- Measure pulse rate accurately.
- Present results in a bar chart.
- Explain a pattern using body systems.

You will need

a stopwatch or clock with a second hand, space to exercise safely

Remember

Find your pulse on the inside of your wrist, below your thumb, using two fingers (not your thumb). Count the beats for **15 seconds**, then multiply by 4 to get **beats per minute**.

1 **Practise the maths. Work out each pulse rate in beats per minute.** [3]

Beats counted in 15 s	× 4	Beats per minute
20	× 4	80
23	× 4	92
31	× 4	124

2 **Arif measured his pulse after different activities. He measured each one three times.**

Activity (1 minute)	Test 1	Test 2	Test 3	Average
sitting still	80	76	78	78
walking	96	100	98	98
jogging	120	124	122	122
skipping fast	148	152	150	150

a) Calculate the averages and write them in the table.

b) Why did Arif repeat each measurement three times?

To make the results more reliable and to check for mistakes.

3 **Draw a bar chart of the average pulse rates.** [4]



4 **Describe the pattern and explain it.** [2]

The harder the exercise, the faster the pulse. Muscles need more oxygen and energy, so the heart pumps faster to deliver more blood.

5 **What could Arif investigate next? Write a question he could test.** [1]

e.g. How long does it take for my pulse to return to normal after exercise? Do fitter people recover faster?

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · A1 · BODY SYSTEMS

Dhaka's air and our lungs

★★★
Grade 5

Learning goals

- Use data to judge air quality.
- Explain how air pollution affects body systems.
- Consider different points of view and suggest actions.

Science | Health | Maths | Geography | Citizenship

Dhaka is one of the busiest cities in the world, and in winter it is often listed among the cities with the **most polluted air**. Smoke from brick kilns, exhaust from vehicles, dust from building sites and burning rubbish add tiny particles to the air. The smallest, called **PM2.5**, are about 30 times thinner than a human hair. They travel deep into the lungs and some even pass into the blood.

Doctors say that breathing polluted air for a long time can cause coughing, asthma attacks and lung infections, and can strain the heart. Children breathe faster than adults, so they take in more polluted air for their size.

Scientists use an **Air Quality Index (AQI)** to tell people how clean the air is.

AQI categories follow the widely used US EPA scale.

AQI	0–50	51–100	101–150	151–200	201–300	301+
Air quality	Good	Moderate	Unhealthy for sensitive groups	Unhealthy	Very unhealthy	Hazardous

1 **A school in Dhaka recorded the AQI each morning for one week in January.** [4]

Day	Sun	Mon	Tue	Wed	Thu
AQI	168	212	245	186	140

a) On which day was the air the most polluted? What was its category?

Tuesday (245): very unhealthy.

b) On how many days was the air "unhealthy" or worse (151 or more)?

4 days (Sun, Mon, Tue, Wed).

c) Suggest why the head teacher might cancel outdoor sports on Tuesday.

During exercise children breathe faster and deeper, taking more polluted air into their lungs.

2 **Which two organ systems are harmed most by air pollution? Explain.** [2]

Respiratory system: particles damage the lungs and airways. Circulatory system: tiny particles enter the blood and strain the heart.

3 **Complete the table to show different points of view.** [3]

Person	What they might say
brick kiln owner	My business gives people jobs; cleaner kilns cost money.
a parent of a child with asthma	The smoke makes my child ill; we need cleaner air now.
a city traffic officer	Old vehicles make a lot of smoke; we need better buses and checks.

4 **Suggest three actions that could reduce air pollution or protect people's lungs.** [3]

e.g. cleaner (modern) brick kilns or alternative blocks; check vehicles for smoke; plant trees; cover building sites and spray water on dust; stop burning rubbish; wear a mask on very polluted days.

ANSWER KEY

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REVIEW AND CHALLENGE · A1 · BODY SYSTEMS

Show what you know

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Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

S	T	O	M	A	C	H	H	F	S	C
O	L	U	N	G	S	N	P	B	J	B
I	N	E	R	V	E	W	J	B	N	H
F	M	K	I	D	N	E	Y	C	K	B
V	E	E	P	B	I	E	V	R	P	O
P	A	G	L	E	B	L	L	U	U	N
H	E	A	R	T	B	L	I	N	L	E
P	T	P	J	E	S	R	O	V	S	T
I	E	A	M	W	E	Y	A	O	E	A
I	T	K	V	H	T	N	I	I	D	R
W	M	E	E	E	D	M	D	F	N	A

blood **bone** **brain** **heart** **kidney** **liver** **lungs** **nerve** **pulse** **stomach**

2 **Which organ system includes the windpipe and the lungs?** [1]

- ○ digestive
- ● **respiratory**
- ○ nervous
- ○ skeletal

3 **Which organ filters the blood to make urine?** [1]

- ○ liver
- ○ bladder
- ● **kidney**
- ○ heart

4 **Sort the organs into the correct systems.** [4]

SORT THESE

spinal cord **heart** **small intestine** **stomach** **nerves** **oesophagus** **brain** **blood vessels**

Digestive	Circulatory	Nervous
stomach	heart	brain
small intestine	blood vessels	spinal cord
oesophagus		nerves

5 **Challenge. A person can live with one kidney but not without *a* heart. Explain why.** [2]

One kidney can filter enough blood on its own, but nothing else can pump blood; without the heart, oxygen and nutrients would not reach the cells.